NEURIPS 2026 WORKSHOP · FLMSEC

System-Prompt Anchoring with Cross-Attention Layers

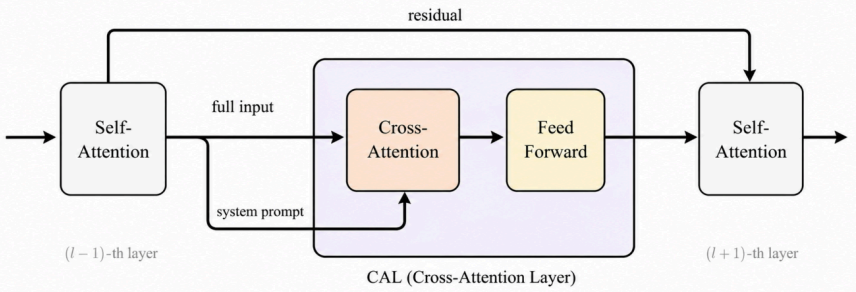
Lixing Li · Cornell University

1 PROBLEM

- Causal self-attention treats privileged system-prompt tokens and untrusted user text with equal priority.
- This leads to **prompt injection** and **context drift**: system rules erode over long or multi-turn inputs.
- Prior defenses rely on preference optimization, structured queries, or hierarchy training rather than a dedicated route.

Question: does a dedicated cross-attention route to the system prompt help, and where in the network should it go?

2 METHOD



A CAL block: queries from the full residual stream, keys/values only from the system-prompt span.

- Zero-initialized CAL blocks trained on a **frozen** decoder backbone (50K-example SFT corpus).
- Ten-placement sweep on Qwen2.5-1.5B; later placements are generally stronger and more parameter-efficient.
- Selected LATE8TH (5 blocks) transferred to Llama-3.1-8B vs. parameter-matched LoRA and ParallelMLP.

3 KEY RESULTS

+7.4 pp
IFEval at 8B vs. matched ParallelMLP (69.13 vs. 61.74, $p<0.05$)

+16.3 pp
SysBench at 8B vs. matched ParallelMLP (58.60 vs. 42.32, $p<0.01$)

38.05 → 25.00%
Many-shot jailbreak attack success at 8B, ParallelMLP → CAL (lower is better, $p<0.01$)

Gains are task-dependent: general ability is largely preserved, but InjecAgent favors ParallelMLP.